

JIANRU DUAN

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EMPLOYMENT	University of Oxford <i>Postdoctoral Research Associate</i> Supervisor: Prof. Dawid Kielak	Oxford, United Kingdom 2026 – 2029
EDUCATION	Peking University <i>Ph.D. Mathematics</i> - Advisor: Prof. Yi Liu Peking University <i>B.S. Mathematics, Huaixin Bachelor</i> - Awarded to top 5% students in the School of Mathematical Sciences at Peking University.	Beijing, China 2021 – 2026 (<i>expected</i>) Beijing, China 2017 – 2021
ACADEMIC VISITS	University of California <i>Visiting Scholar</i> Host: Prof. Ian Agol	Berkeley, CA, United States Oct. 2024 – Apr. 2025
RESEARCH INTERESTS	Topology of 3-manifolds L^2 -theory and geometric group theory	
PREPRINTS AND PUBLICATIONS	- Flexible exponent of geometric 3-manifolds and Legendrian maps of Seifert spaces (with Jianfeng Lin, Shicheng Wang, Zhongzi Wang and Dongyi Wei). Preprint available at https://arxiv.org/abs/2605.17257. - Universal L^2-torsion and sutured decomposition for 3-manifolds. Preprint available at https://arxiv.org/abs/2512.01305. - Guts determine the leading coefficients of L^2-Alexander torsions. <i>Trans. Amer. Math. Soc.</i> 378 (2025), 3699–3720. - On the positivity of twisted L^2-torsion for 3-manifolds. <i>Algebr. Geom. Topol.</i> 24 (2024), 2307–2329.	

AWARDS AND HONORS	• BICMR Mathematical Award for Graduate Students	2025–2026
	• Peking University President’s Scholarship	2023–2025
	• National Scholarship for Graduate Student	2023
	• Huaixin Bachelor (BICMR)	2021
	• Peking University President’s Scholarship	2021
	• Outstanding Student Award of Beijing	2020
	• National Scholarship for Undergraduate Student	2018
INVITED TALKS	• The 21st East Asian Conference on Geometric Topology at Ewha Womans University, Seoul, February 2026.	
	• The 11th Sino-Russian Conference on Knot Theory and Related Topics at Xi’an Jiaotong-Liverpool University, Suzhou, July 2025.	
	• Geometric Group Theory Seminar at Shanghai Center for Mathematical Sciences, Fudan University, May 2025.	
	• Caltech Geometry & Topology Seminar, April 2025.	
	• Rutgers Geometry & Topology Seminar, February 2025.	
	• Topology Seminar at UC Berkeley, January 2025.	
	• Gave a talk at South China University of Technology, Guangzhou, June 2024.	
	• YMSC Topology Seminar at YMSC, Tsinghua University, April 2024.	
ACADEMIC SERVICES	Teaching assistant of Homology Theory in 2021 Spring	
	Teaching assistant of Low Dimensional Topology 2022 Fall	
	Teaching assistant Homology Theory in 2024 Spring	
	Student editor of Front. Math., 2023–2024	